

CS 4403 - DATA MINING

HUMAN VS AI

A DATA MINING STUDY OF GENERATED VS AUTHENTIC DATASETS

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PROBLEM

- AI-generated text is increasing rapidly
- Hard to distinguish real vs synthetic speech
- Need automated detection

PROJECT GOAL

1

Analyze speech
datasets

2

Compare real vs
synthetic text

3

Train models to classify
them

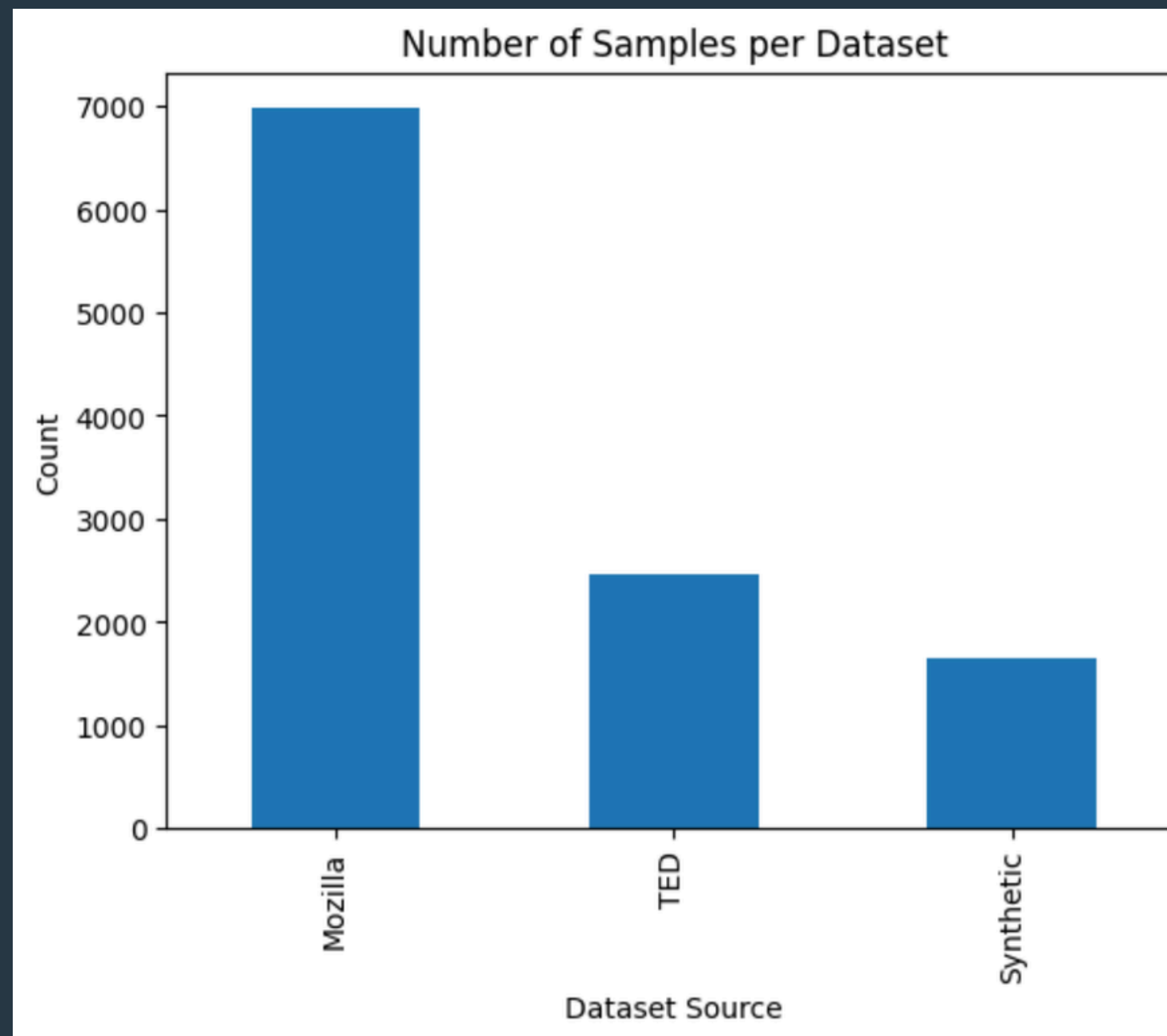
DATASETS USED

- TED Talks - real speeches
- Mozilla Common Voice -short spoken text
- Synthetic dataset - AI-generated

DATASET DISTRIBUTION

Overview

- 11,000 samples after cleaning
- Mozilla is the largest dataset, followed by TED and synthetic
- Datasets were balanced



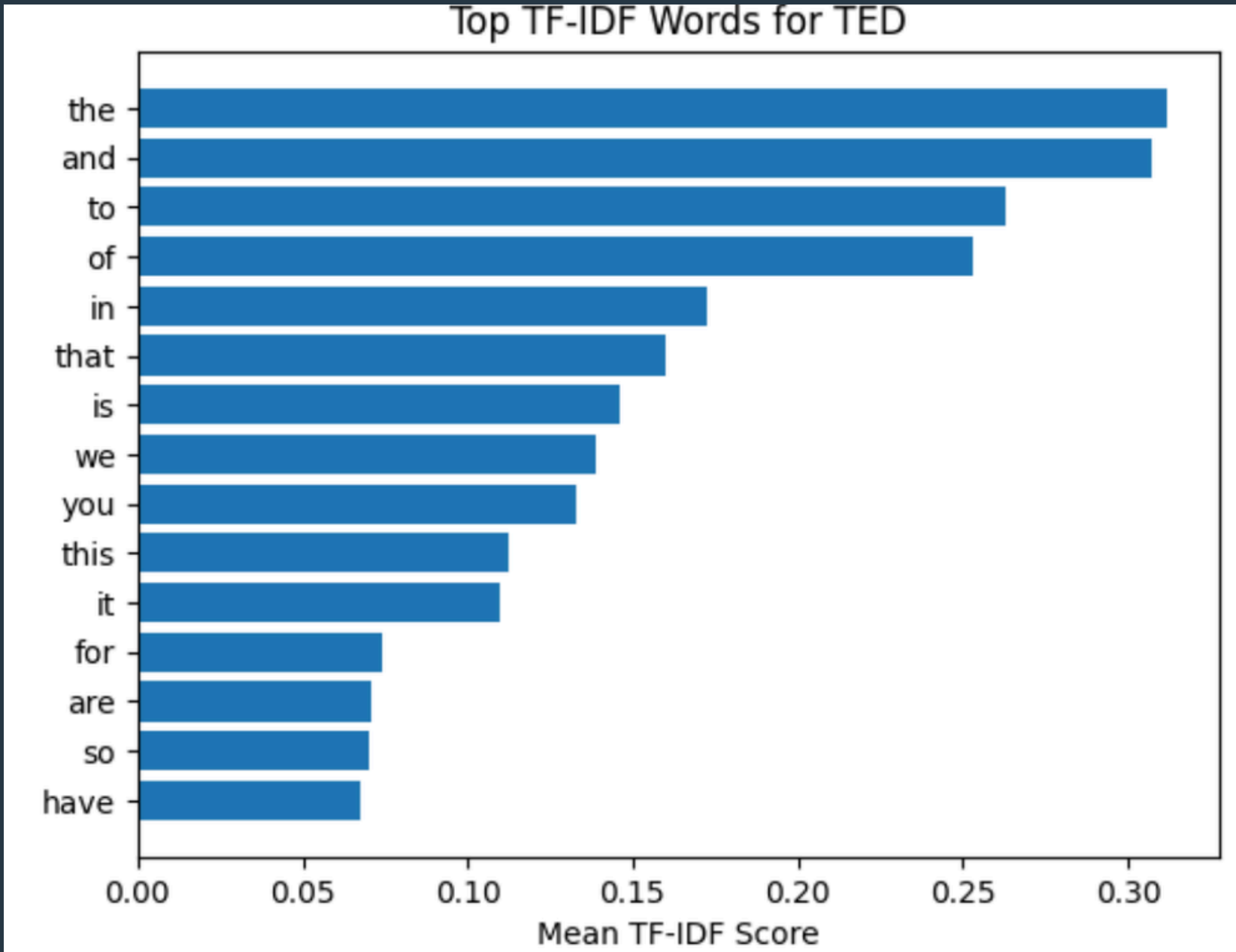
DATA PREPROCESSING

- Remove missing & duplicates
- Lowercase text
- Remove punctuation
- Remove short sentences

OUR APPROACH

- TF-IDF
- Train/ test split
- Models used:
 - Naive Bayes
 - Logistic Regression
 - SVM
 - Random Forest

KEY PATTERN IN DATA



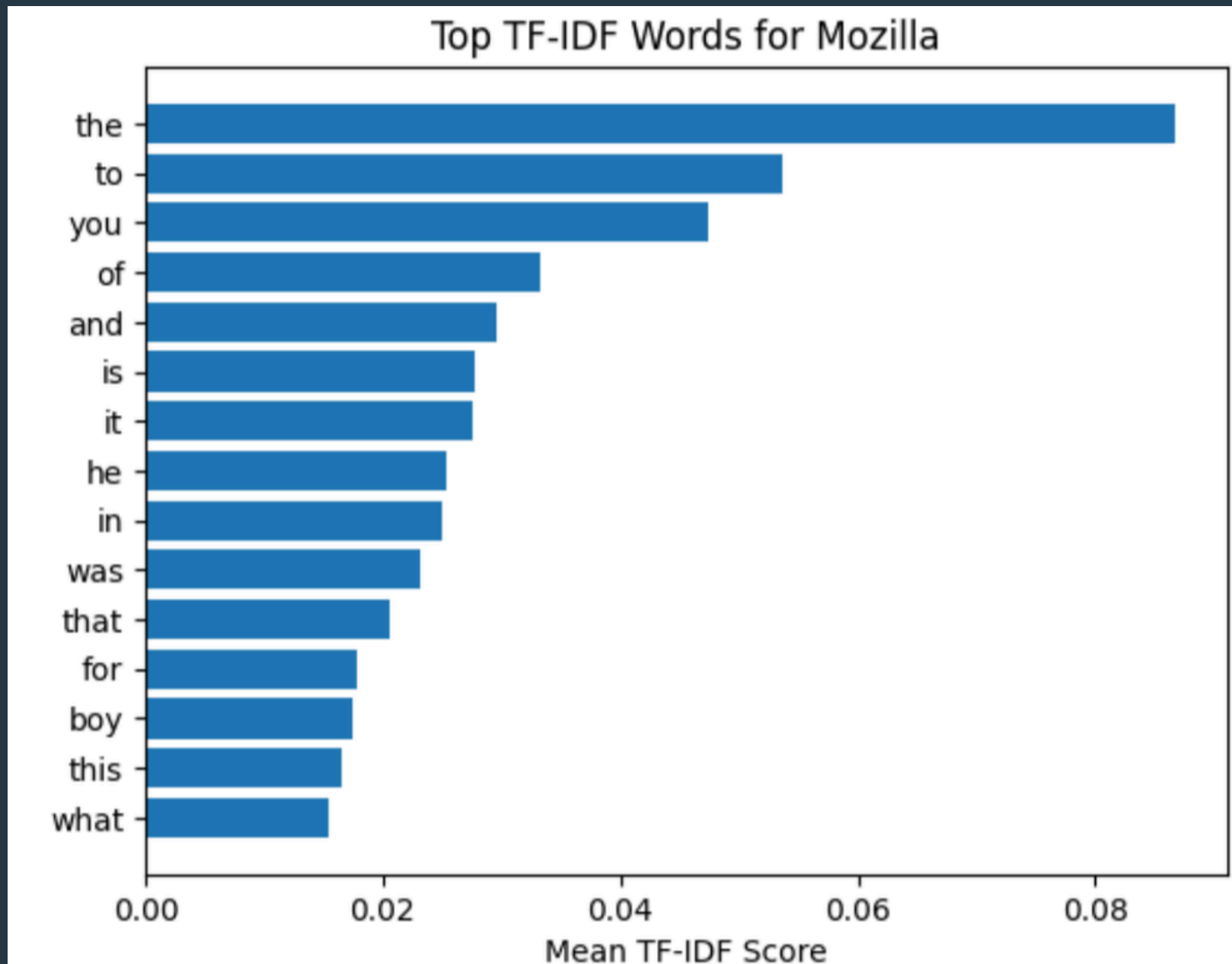
TED Dataset

This dataset uses conversational words

Most used word is “the”

Least used word is “have”

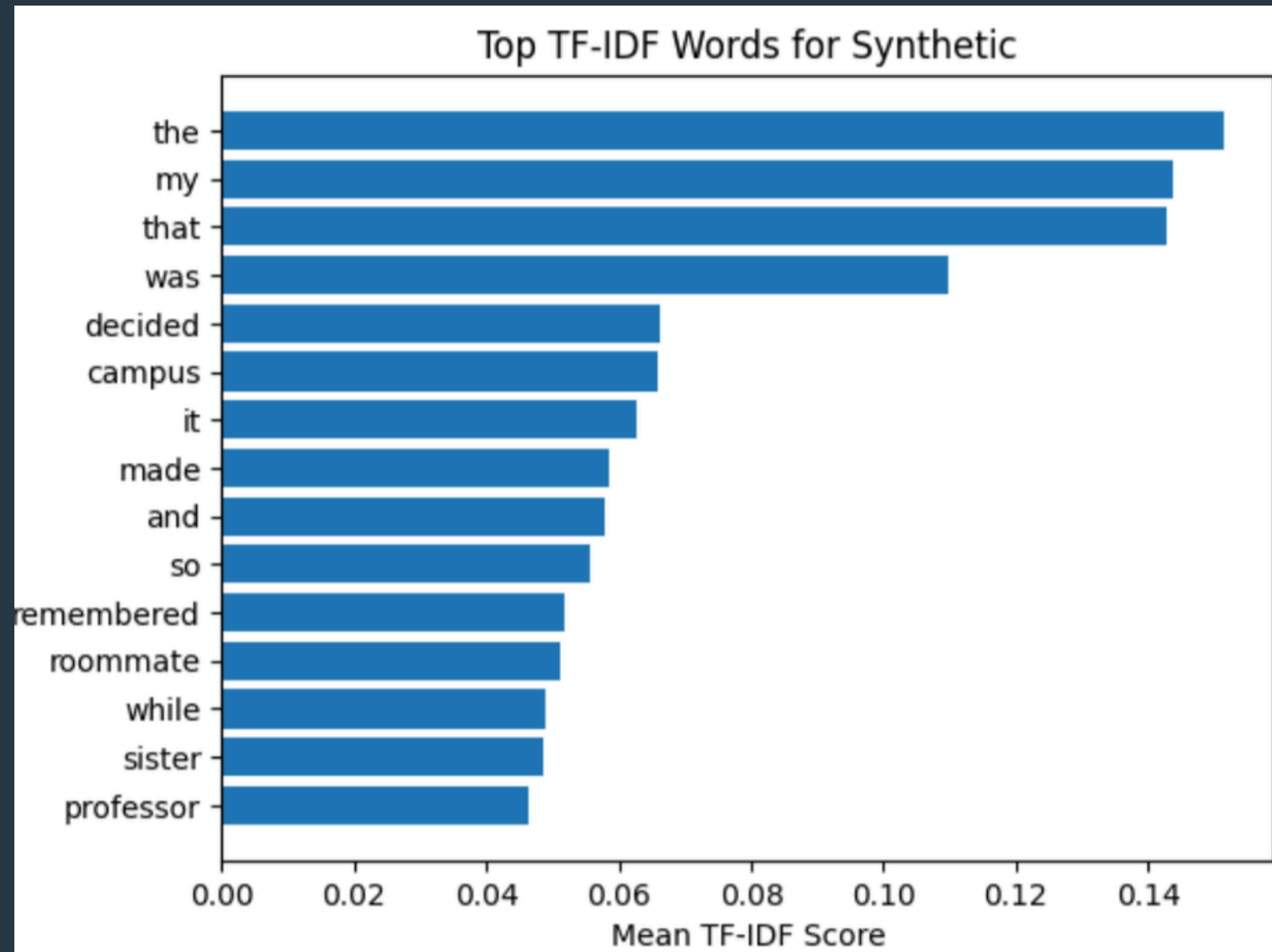
KEY PATTERN IN DATA



Mozilla Dataset

- This dataset is for simple sentences
- Most used word is “the”
- Least used word is “what”

KEY PATTERN IN DATA



Synthetic Dataset

- This dataset is for themed words
- Most used word is “the”
- Least used word is “professor”

MODEL PERFORMANCE

Model	Accuracy
Naive Bayes	95.9%
Logistic Regression	99.7%
SVM	99.8%
Random Forest	99.7%

PERFORMANCE INSIGHTS

- Very few errors
- Synthetic data easiest to detect
- Some confusion between TED and Mozilla

KEY INSIGHTS

- AI-generated text has identifiable patterns
- Text length differs across datasets
- ML models can detect synthetic text accuracy, however results are influenced by dataset differences.

LIMITATIONS

- Overly High Accuracy
- Synthetic dataset may not represent all types of AI-generated text
- TF-IDF only captures word frequency
- Models may learn dataset-specific patterns instead of true differences

EXPLORATORY DATA ANALYSIS (EDA)

Understanding why our model achieved unusually high accuracy

- Feature Engineering and Testing
- Cross-Domain Validation
- Root cause Investigation of High Accuracy

WHAT WE LEARNED

- Very high accuracy can sometimes be misleading.
- Models may learn dataset characteristics rather than true patterns.
- Structural differences can dominate results.
- Multiple experiments are needed to validate model performance.
- Careful dataset design is essential for meaningful AI evaluation.
- Cross-domain accuracy dropped to ~19%, showing poor generalization



THANK YOU

FOR YOUR ATTENTION